4.5 Nuclear and Particle Physics

This topic covers atomic structure, particle accelerators and the standard quark-lepton model.

This topic is the subject of current research, involving the acceleration and detection of high-energy particles. It may be taught by exploring a range of experiments such as

- alpha scattering and the nuclear model of the atom
- accelerating particles to high energy
- detecting and interpreting interactions between particles.

Candidates will be assessed on their ability to:

111	understand what is meant by *nucleon number (mass number)* and *proton number (atomic number)*
112	understand how large-angle alpha particle scattering gives evidence for a nuclear model of the atom and how our understanding of atomic structure has changed over time
113	understand that electrons are released in the process of thermionic emission and how they can be accelerated by electric and magnetic fields
114	understand the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)
115	be able to derive and use the equation $r = \frac{p}{BQ}$ for a charged particle in a magnetic field
116	be able to apply conservation of charge, energy and momentum to interactions between particles and interpret particle tracks
117	understand why high energies are required to investigate the structure of nucleons
118	be able to use the equation $\Delta E = c^2 \Delta m$ in situations involving the creation and annihilation of matter and antimatter particles
119	be able to use MeV and GeV (energy) and MeV/c^2, GeV/c^2 (mass) and convert between these and SI units
120	understand situations in which the relativistic increase in particle lifetime is significant (use of relativistic equations not required)
121	know that in the standard quark-lepton model particles can be classified as: • baryons (e.g. neutrons and protons), which are made from three quarks • mesons (e.g. pions), which are made from a quark and an antiquark • leptons (e.g. electrons and neutrinos), which are fundamental particles • photons and that the symmetry of the model predicted the top quark
122	know that every particle has a corresponding antiparticle and be able to use the properties of a particle to deduce the properties of its antiparticle and vice versa